
MODULE *Exokernel*

EXTENDS *Naturals, Sequences*

Parameters and constants

CONSTANT *TaskCount, ResourceCount*

VARIABLE *tasks, resources, allocations*

TaskId are modeled as a sequence of task *IDs*, where each task has an index.

$TaskId \triangleq 1 \dots TaskCount$

ResourceId are modeled as a sequence of resource *IDs*, where each resource has an index.

$ResourceId \triangleq 1 \dots ResourceCount$

$TaskState \triangleq \{ \text{"waiting"}, \text{"running"} \}$

$Availability \triangleq \{ \text{"free"}, \text{"busy"} \}$

Type invariant of the specification

$ExokernelTypeInvariant \triangleq$
 $\wedge tasks \in [TaskId \rightarrow TaskState]$
 $\wedge resources \in [ResourceId \rightarrow Availability]$
 $\wedge allocations \in [TaskId \rightarrow [ResourceId \rightarrow \text{BOOLEAN}]]$

Define *Exokernel* state as a record containing tasks, resources, and allocations

$ExokernelInit \triangleq$
 $\wedge tasks = [t \in TaskId \mapsto \text{"waiting"}]$
 $\wedge resources = [r \in ResourceId \mapsto \text{"free"}]$
 $\wedge allocations = [t \in TaskId \mapsto [r \in ResourceId \mapsto \text{FALSE}]]$

A task can request access to a resource if it's not already in use.

$RequestResource(t, r) \triangleq$
 $\wedge resources[r] = \text{"free"}$
 $\wedge allocations[t][r] = \text{FALSE}$
 $\wedge tasks[t] = \text{"waiting"}$
 $\wedge tasks' = [tasks \text{ EXCEPT } !t = \text{"running"}]$
 $\wedge resources' = [resources \text{ EXCEPT } !r = \text{"busy"}]$
 $\wedge allocations' = [allocations \text{ EXCEPT } !t[r] = \text{TRUE}]$

$FreeResource \triangleq \text{CHOOSE } r \in ResourceId : resources[r] = \text{"free"}$

$OccupyResource \triangleq$

LET $res \triangleq \text{CHOOSE } r \in ResourceId : resources[r] = \text{"free"}$
 IN $resources[res] = \text{"busy"}$

A task can release a resource when it has finished using it.

$ReleaseResource(t, r) \triangleq$
 $\wedge allocations[t][r] = \text{TRUE}$

$$\begin{aligned}
&\wedge tasks[t] = \text{"running"} \\
&\wedge tasks' = [tasks \text{ EXCEPT } ![t] = \text{"waiting"}] \\
&\wedge resources' = [resources \text{ EXCEPT } ![r] = \text{"free"}] \\
&\wedge allocations' = [allocations \text{ EXCEPT } ![t][r] = \text{FALSE}]
\end{aligned}$$

The next-state relation defines valid state transitions

$$\begin{aligned}
ExokernelNext &\triangleq \\
&\exists t \in TaskId, r \in ResourceId : \\
&\quad (RequestResource(t, r) \vee ReleaseResource(t, r))
\end{aligned}$$

Specification of the overall system behavior

$$ExokernelSpec \triangleq ExokernelInit \wedge \Box [ExokernelNext]_{\langle tasks, resources, allocations \rangle}$$

$$vars \triangleq \langle exokernel \rangle$$

$$Fairness \triangleq WF_vars(TaskAction)$$

Invariant: No two tasks can hold the same resource simultaneously

$$\begin{aligned}
ResourceExclusivity &\triangleq \\
&\forall r \in ResourceId : \\
&\quad \forall t1, t2 \in TaskId : (t1 \neq t2) \Rightarrow \neg(allocations[t1][r] \wedge allocations[t2][r])
\end{aligned}$$

$$ASSUME \text{ Assumption} \triangleq tasks \in [TaskId \rightarrow TaskState]$$

THEOREM $Init \Rightarrow Fairness$

BY DEF $Init, Fairness$